


# Introduction to Fink broker

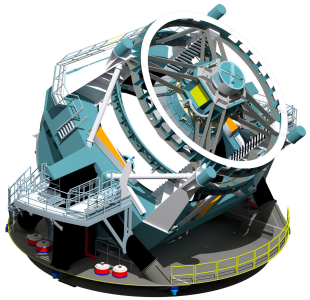
Emille Ishida, Julien Peloton and Anais Möller

*Rubin-LSST Alert Brokers: alerts to discoveries - IUCAA, 2026*

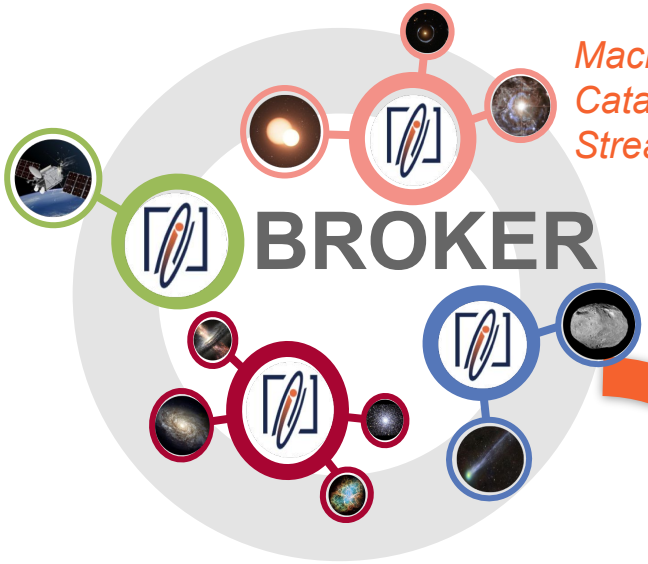
# The data path



*every ~30 seconds down to mag ~24*



*10 million alerts per night…*




*Machine learning*
*Catalog association*
*Streams join*





*We would like the **interesting** ones ...*

# Brokers as discovery machines



Alert information solely is not enough – we need smooth **communication protocols** with experts and follow-up facilities

- Enable users to add features to the broker
- Keep a communication channel open for constant update/feedback
- Machine learning tools should adapt to different science cases

# Philosophy

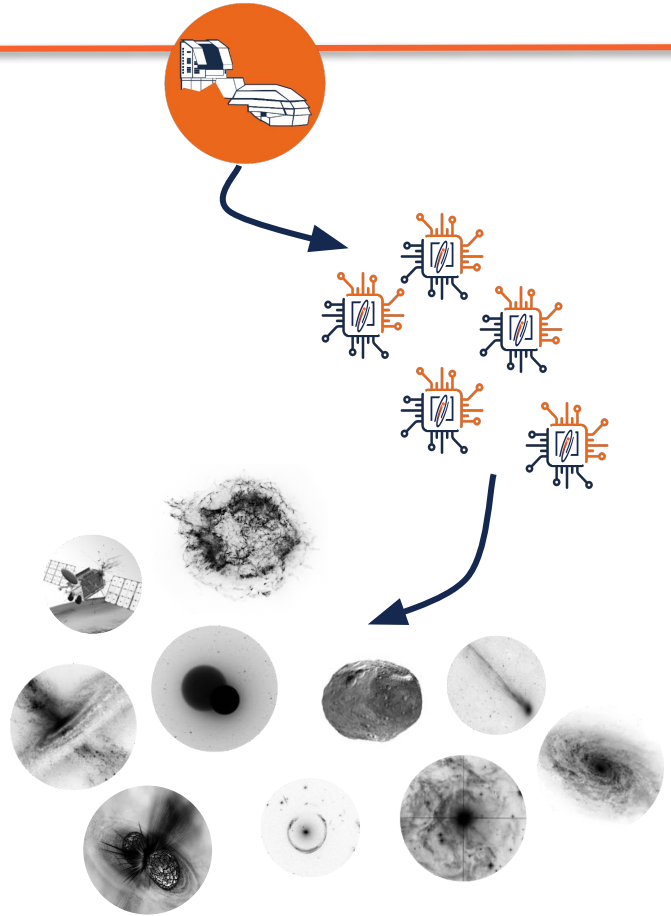
Centralised processing, decentralised science

- Fink core team provides infrastructure & technical assistance, users extend Fink capabilities by providing scientific codes

At the core of Fink are the *science modules*

- Enrichment of the alert packet. Output is available to anyone.
- Divide and conquer.
- One man's trash is another man's treasure
- 14 science modules currently

Each Fink community project is responsible for defining its own publication policy



## Supernovae

*Allam+2023, Leoni+2022, Russeil+2026, Möller+2025/2022b*
*CNRS MITI grant*

## Nuclear transients

*Russeil+2022*
*Llamas-Lanza+ 2025*
*Quintin+ 2025*
*Hamo+ 2025*

*Swinburne-CNRS IEA grant*

## Solar System

*Le Montagner+ 2023*
*Carry+2024*
*CNRS MITI grant*

## Magnetic CVs

*Lima, Rodrigues+ in prep.*

## YSO

*Kuhn+ in prep.*

## LC representation

*Gondhalekar+ 2026*

## LC visualisation

*Steinwender+ 2026*

## LSST multi-class

*Fraga+ 2024*

FINK

Your science case here!

https://fink-broker.org/papers

## MMA

*Astrocolibri*

## Kilonovae

*Grandma+Fink 2022,2023*
*Biswas+2022,2023*
*Stevenson+ 2026*

## Gamma Ray Bursts

*Biteau+ 2026*

## Hostless

*Pessi+ 2024*
*R. Durgesh 2026*

## Space Awareness

*Karpov+2023,2022*

## Anomalies

*Pruzhinskaya+ 2026*

## Exoplanets

*Ban+ 2025*

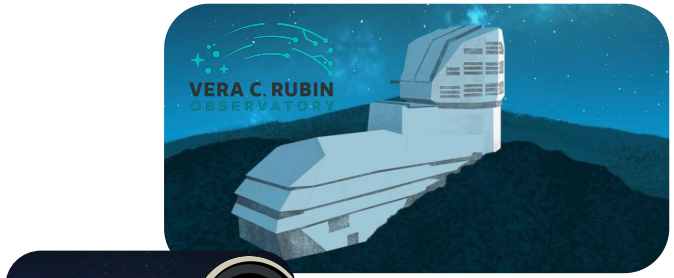
# Consequence:

# Broad spectrum of applications

# Fink in a nutshell

https://fink-broker.org

Fink is a broker serving the scientific community by ingesting, classifying, filtering, and redistributing alerts from telescopes and surveys.



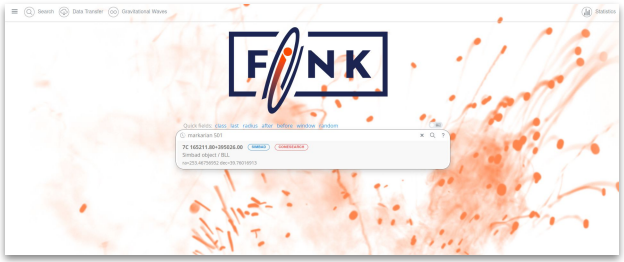




As of 2026/06: +150 collaborators, 26 countries

Services deployed on large OpenStack clouds (UPSaclay & CC-IN2P3)

- Scalable to millions of alerts per night




Operating 24/7 since 2019, serving 100+ unique users per day (scientists, follow-up facilities & amateurs)

# Important distinction

## Science Module

Python function:

***f(alert; ++) ⇒ boolean or float***

Can be as complicated as you want, but it should be computationally efficient

Impact on the alert package:

Add value (**a new column** at the schema)

Threshold is defined by the user:

Does not affect the number of alerts

Currently available:

- Xmatch
  *SIMBAD, Gaia, SPICY, Mangrove, VSX, 3HSP, 4LAC*
- Statistical features: light_curve by Malanchev *et al.*, 2021
- Anomaly score - *Pruzhinskaya, Kornilov (MSU)*
- Early SNIa - *Leoni et al., 2021, A&A*
- SN & SNIa - *Möller and de Boissiere, 2020, MNRAS*
- Kilonova - *Biswas et al., 2023, A&A*
- Microlensing - *Bachelet, Makler et al.,*
- Hostless - *Durgesh et al., 2026, Pessi et al. 2025*

ZTF: *https://doc.ztf.fink-broker.org/broker/science_modules/*
LSST*:* *https://doc.lsst.fink-broker.org/science/science_modules/*

# Important distinction

Currently available:

- Early SNIa candidates
- TNS xmatch
- SPICY rising
- Magnetic CVs
- SN candidates
- Early TDE candidates
- Hostless candidates
  - Intra-night hostless
  - Extra-night hostless
- Solar System Objects candidates
- Kilonova candidates
  - ML, rate based, xmatched based
- GRB
- Gravitational Waves

  + after night filters

## Filter

**Selection cuts** upon existing columns;

It can use output from any science module.

Impact on the alert package:

Decrease the number of alerts redirected to the user

How to access it:

Kafka stream

Bots (telegram, slack, +)

ZTF: https://doc.ztf.fink-broker.org/broker/filters/

LSST: https://doc.lsst.fink-broker.org/science/filters/

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All of them exist because someone **asked** for it !

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All of them exist because someone **asked** for it !

All outputs are public and accessible to everyone

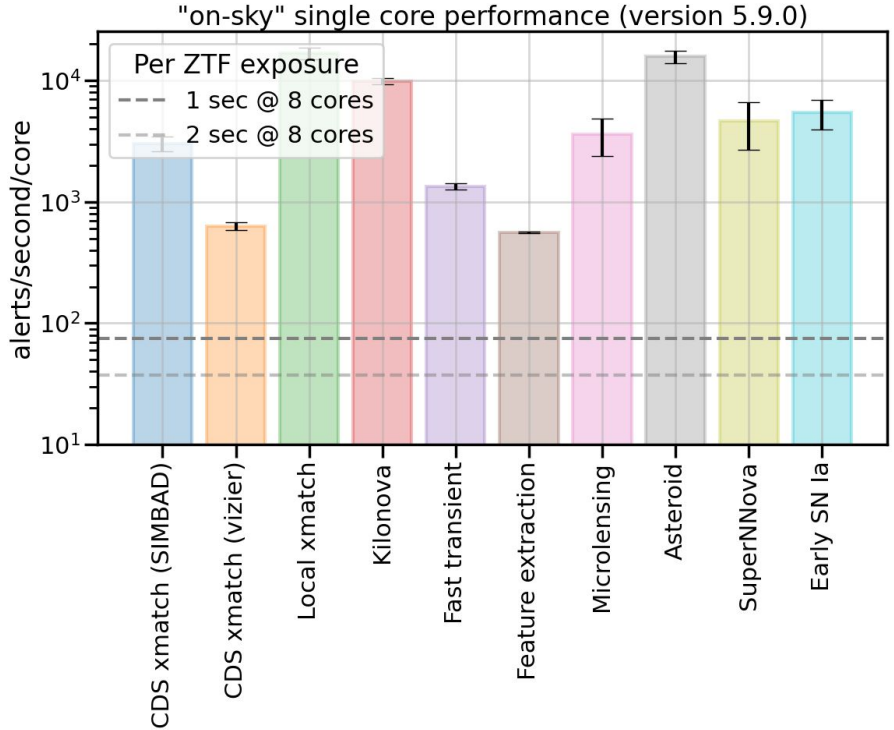
# Science modules

Fink has a community-driven roadmap

We currently host 12+ science modules:

- Simple feature-based (decision tree, fit, etc.)
- Crossmatch with external catalogs
- ML-based

They are all tested, profiled, curated, modified as dependencies change (67 Python deps!).



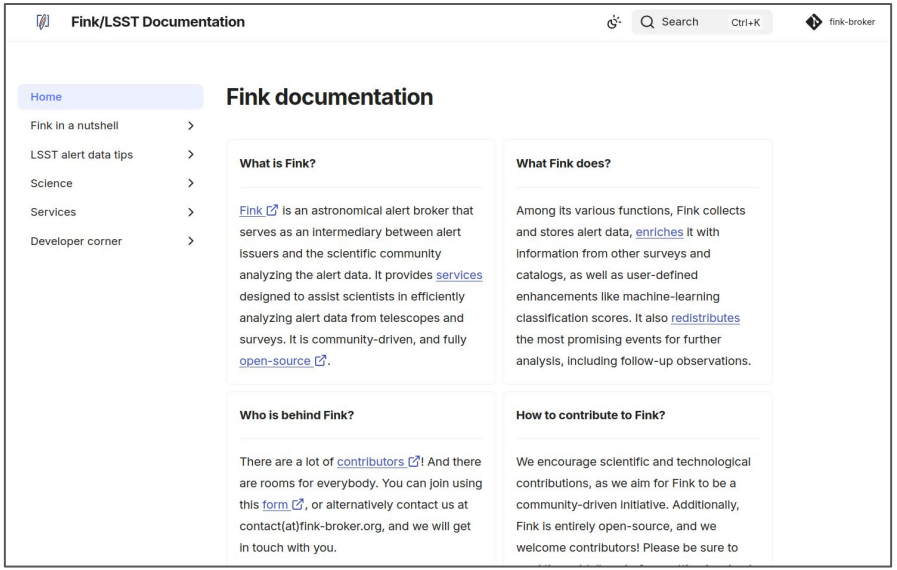
# Documentation

Extensive documentation available online

- How the broker works
- What is available (data, schema, …)
- How to use the different services

Probably incomplete, or outdated!

- Reach us if you find missing or wrong information



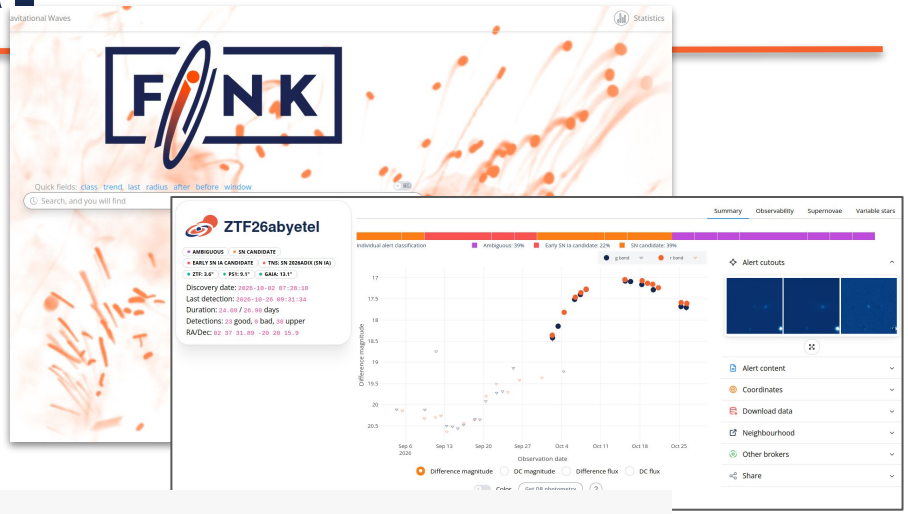

ZTF: https://doc.ztf.fink-broker.org/
LSST: https://doc.lsst.fink-broker.org/

# Science Portal

https://ztf.fink-portal.org/
https://lsst.fink-portal.org/

**Goal:** Enable simple queries on name, position or alert class

- Access to 200M+ alerts from ZTF from the **Fink database**
- Let you **visualise** and **interact** with the results on the browser
- Easy to connect with your portal/online resources
- For more complex analyses, see Data Transfer or Livestream



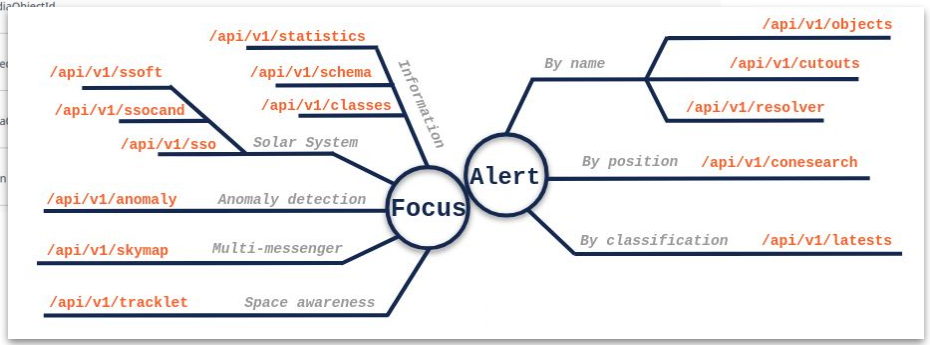



# REST API

https://api.ztf.fink-portal.org/
https://api.lsst.fink-portal.org/

API to access data programmatically, regardless of the programming language you choose.





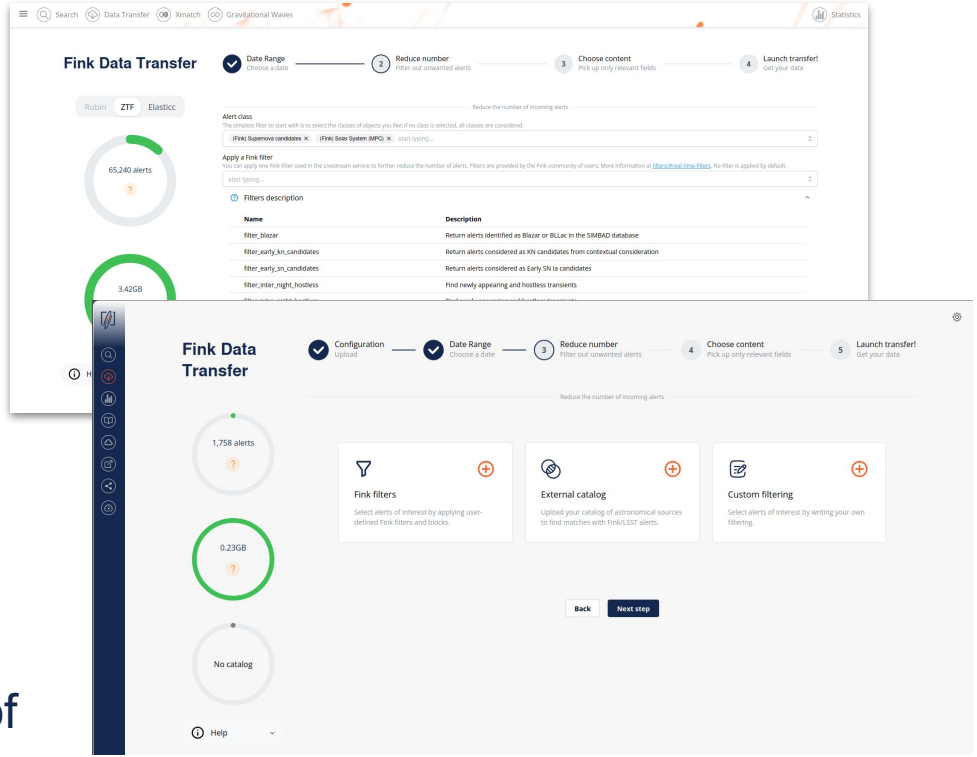
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Science Roadmap
Technological consideration
LSST alert data tips
From ZTF to LSST
Photometry
Alert data quality
Known issues
Science
Fink science modules
Fink filters
Classification
Services
Overview
REST API
Livestream
Fink bots
Data Transfer

# Data Transfer

https://ztf.fink-portal.org/download
https://lsst.fink-portal.org/download

**Goal:** enable complex analyses, on large portions of processed data.

- Access to 200M+ alerts from ZTF from the **Fink Data Lake**
- Each job is allocated **8 vCPUs** on the Spark cluster.
- Customisation: dates, filters, content
- The output data is available for download for 7 days, and can be downloaded an unlimited number of times.

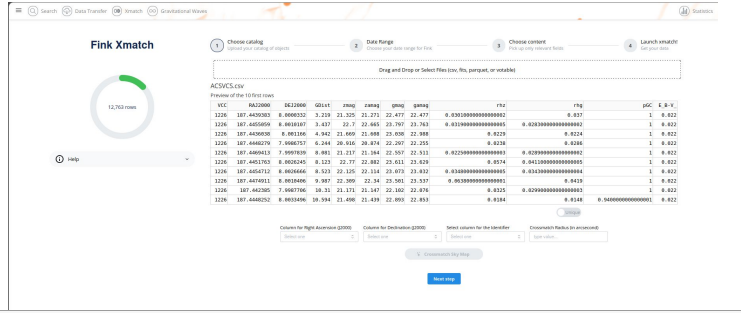


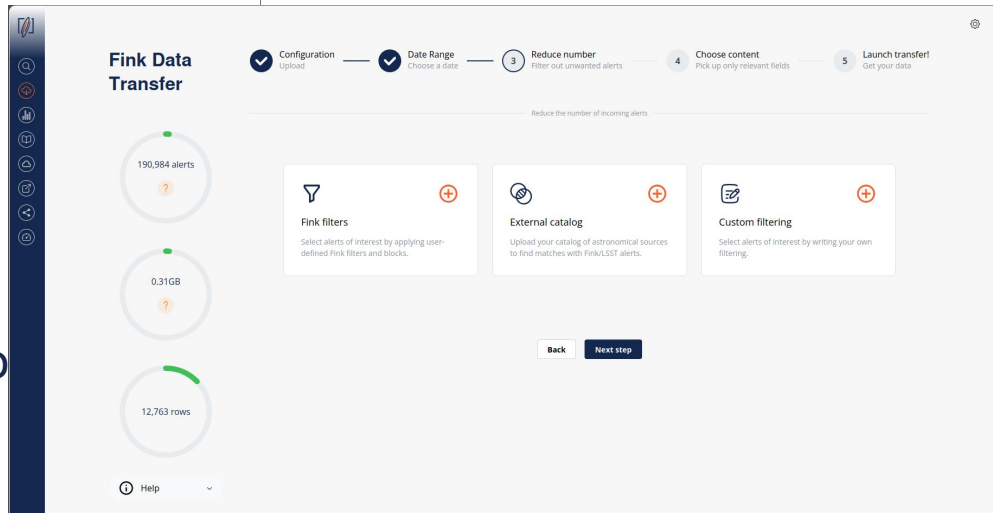
# Xmatch

https://ztf.fink-portal.org/xmatch
https://lsst.fink-portal.org/download

**Goal:** enable large table crossmatch with Fink alert data

- User-defined catalogs with up to 100,000 rows
- Access to 200M+ alerts from the **Fink Data Lake**
- Customisable output schema
- The output data is available for download for 7 days, and can be downloaded an unlimited number o times.



# Livestream & Fink bots

**Goal:** define and retrieve alerts of interest (for *you*)

- Facilitate follow-up observations or heavy analyses
- Rely on fink-filters (doc: ZTF / LSST)
- Output: Kafka stream, Slack/TG channels
- It gives all flexibility to the users about the definition of cuts and filters, and it should be de-facto the primary choice for real-time analyses.

**Fink Bots**

The bots allow you to access alert or object data selected by some Fink filters in real-time or after the night. Information is sent to instant messaging services, such as Slack or Telegram. Best to scroll on a bus! Login can apply depending on the service.

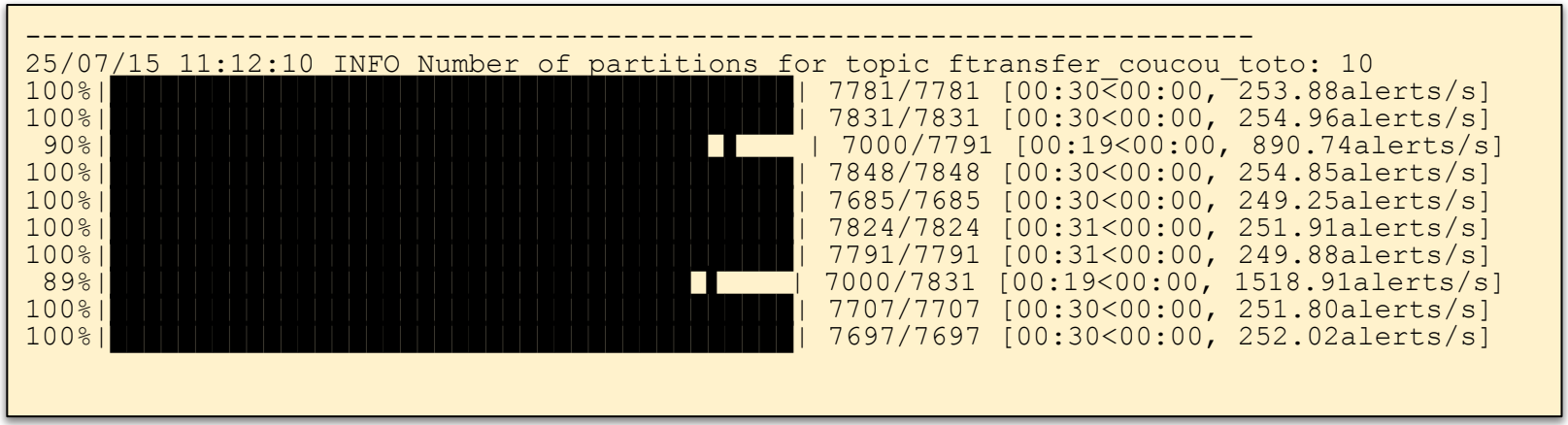


@FINK_EARLY_IA

# Fink client

`fink-client` is a package to manipulate alerts issued from Fink programmatically

- It is a convenient wrapper around Kafka low-level functionalities.
- It is used to retrieve data from the **Data Transfer, Xmatch** (`fink_datatransfer`) & the **Livestream** (`fink_consumer`) services
- Multithreaded to increase the throughput, but sensitive to timeouts.



# Follow-up tools

**Goal:** ease follow-up of targets

Fink Module for **TOM Toolkit**

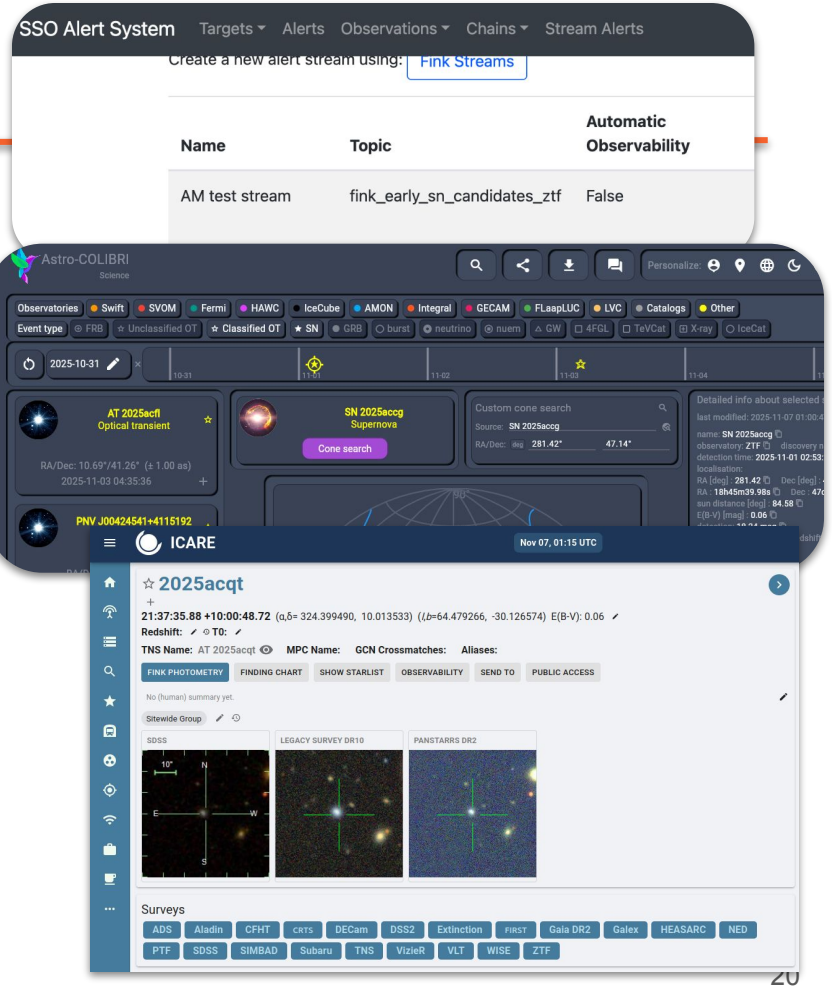
- https://github.com/TOMToolkit/tom_fink
- REST API & livestream modes
- Used by Siding Spring Observatory

**Astro-COLIBRI**

- Send targets (see Julian's work)

**SkyPortal/Icare** Fink Client

- Ingest alerts data from Fink into SkyPortal's database



# Fink services

- Live streams (Kafka streams): fink-client
  - *What for? Live inspection, Follow-up*
  - Personalisable filters to select objects/parameters of interest
  - Data received “live” (+processing delays)
- Science Portal (dash-based) & REST API: ZTF / LSST
  - *What for? Visual inspection, small queries, daily monitoring*
  - All data processed remains accessible
- Data Transfer service: fink-client, post, links: ZTF / LSST
  - *What for? Bulk download, complex queries, ML/DL training, exotic analyses*
- TOM module
  - *Whatfor? Follow-up*
  - https://github.com/TOMToolkit/tom_fink

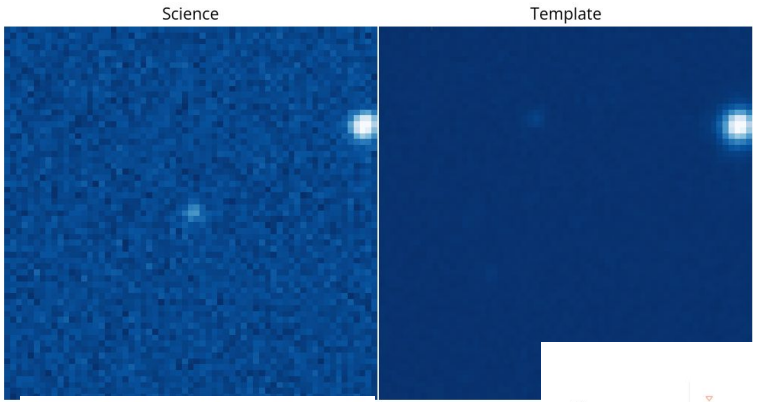


# Development Policy

Our goal is to have a clear science case, **and team**, associated with each filter / science module

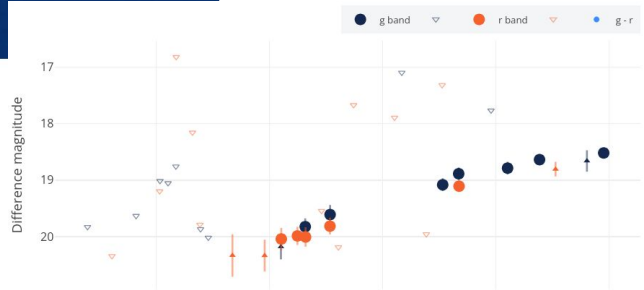
# Example: Hostless transients

ELEPHANT: ExtragaLactic alErt Pipeline for Hostless AstroNomical Transients

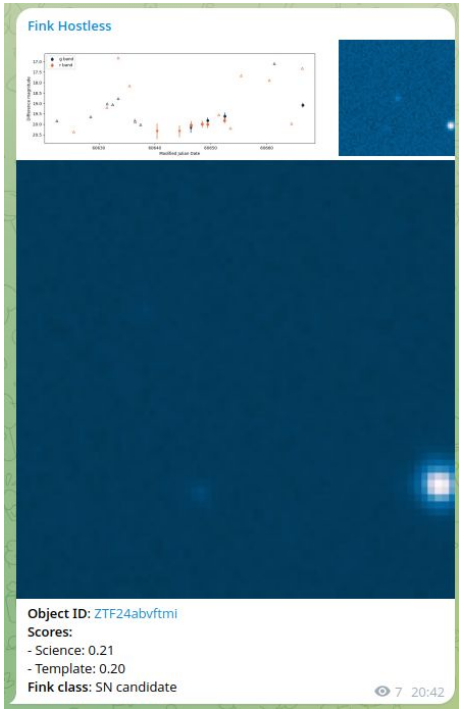



**ZTF24abvftmi**

https://fink-portal.org/ZTF24abvftmi




*Hostless SLSN which appear in the middle of holidays …*




@FINK_HOSTLESS

*Priscila Pessi et al., 2024, Astronomy & Astrophysics, Volume 691, A181*

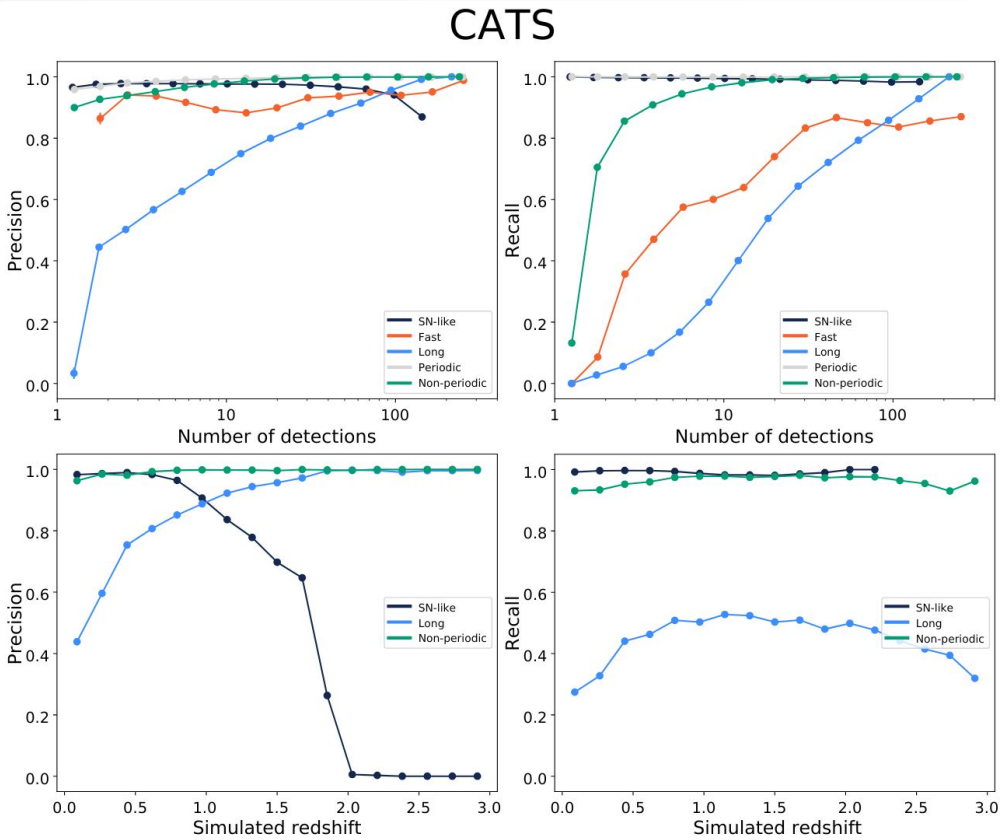
# Example: Multi-class classification

## CATS

*The CBPF alert transient search*

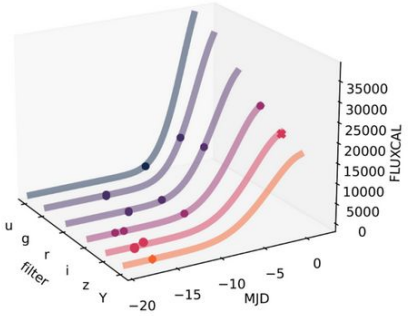
- Built on ELAsTiCC
- Taxonomy of broad classes
- Performance with reduced number of points

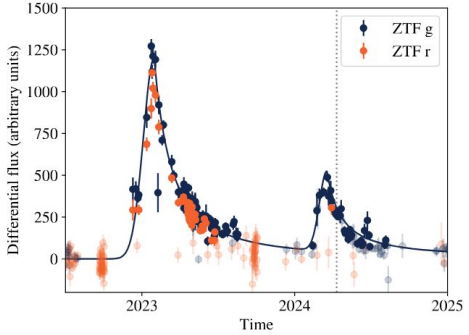
*CATS will act as a primary filter for the most expensive binary classifiers*

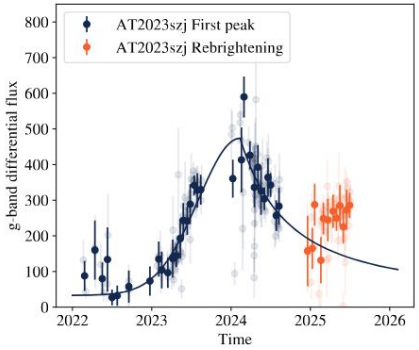



*Fraga et al., 2024, A&A, Volume 692, id.A208*

# Example: Tidal Disruption Events

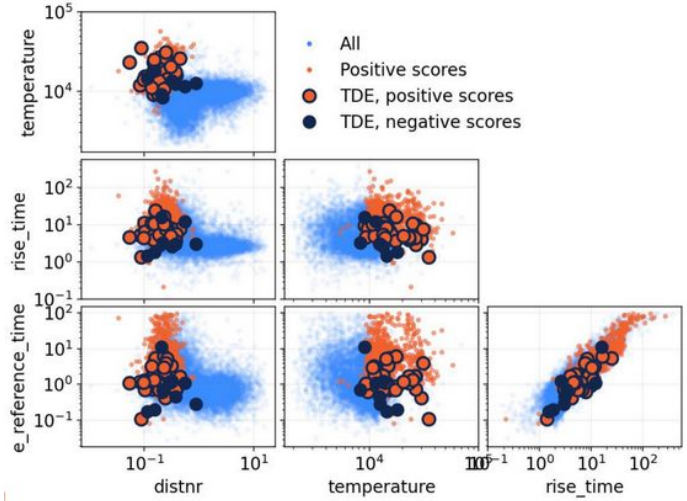









$$F_\nu(t,\nu) = \frac{\pi}{\sigma_{SB}} \times \frac{B_\nu(T,\nu)}{T(t)^4} \times F_{bol}(t)$$

$$T(t) = T_{min} + \frac{T_{max} - T_{min}}{1 + e^{\frac{t - t_0}{\tau}}}$$

$$F_{bol}(t) = \frac{A}{e^{-\frac{t-t_0}{\tau_{rise}}} + e^{\frac{t-t_0}{\tau_{fall}}}}$$

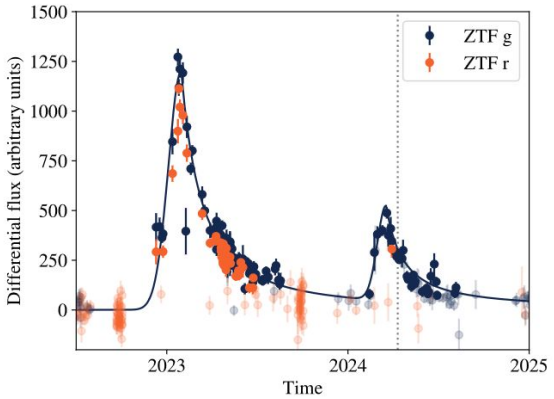
Llamas-Lanza *et al.,* 2025, A&A - submitted - [arXiv:astro-ph/2507.17499](arXiv:astro-ph/2507.17499)
Quintin et al., 2025, A&A - submitted - [arXiv:astro-ph/2511.19016](arXiv:astro-ph/2511.19016)

Case study:

# Repeated TDE candidates

Quintin et al., 2026

| Name | Host | Approx. Interval (d.) | Flares | Peak Evolution | "Next" Peak | Reference |
|---|---|---|---|---|---|---|
| OJ 287 | Active | 4000 | ~10 | Similar | ~2030 | Sillanpaa et al. (1988) |
| IC 3599 | Active | ~3470? | 2/3 | Similar | 2019 (No) | Campana et al. (2015) |
| ASASSN-14ko | Active | 115.2 | ~30 | Similar | Aug. 2025 | Payne et al. (2021) |
| AT 2019avd | Passive | ~500 | 2 | Lower | May 2022 (No) | Malyali et al. (2021) |
| AT 2019azh | Passive | ~4800 | 2 | Similar | 2032 | Hinkle et al. (2021) |
| AT 2018fyk | Passive | ~1200 | 2 | Lower | Apr. 2025 (No) | Wevers et al. (2022) |
| †AT 2019baf | Passive | 250 | 2 | Lower | Jul. 2020 (No) | Yao et al. (2023) |
| AT 2019ehz | Passive | ~300 | 2 | Lower | Dec. 2020 (Hints) | Yao et al. (2023) |
| †AT 2020acka | Passive | 300 | 2 | Lower | Aug. 2022 (No) | Yao et al. (2023) |
| †AT 2021uqv | Passive | 225 | 2 | Lower | Jan 2023 (Hints) | Yao et al. (2023) |
| AT 2020vdq | Passive | ~870 | 2 | Higher | Oct. 2025 | Somalwar et al. (2023) |
| †AT 2021loi | Active | ~400 | 2 | Lower | Aug. 2023 (No) | Makrygianni et al. (2023) |
| IRAS F01004-2237 | Passive | ~4175 | 2 | Similar | 2032 | Makrygianni et al. (2023); Sun et al. (2024) |
| AT 2022dbl | Passive | ~710 | 2 | Lower | Jan. 2026 | Lin et al. (2024) |
| AT 2019aalc | Active | 1500 | 2 | Higher | Aug. 2028 | Milán Veres et al. (2024) |
| AT 2024pvu | Passive | ~6500 | 2 | Similar | 2042 | Langis et al. (2025) |
| †AT 2022exr | Passive | 120-50 | 2/3 | Lower | Nov. 2022 (Gap) | Langis et al. (2025) |
| †AT 2021uvz | Passive | 175 | 2 | Lower | Dec. 2021 (Gap) | Langis et al. (2025) |
| AT 2019teq | Passive | 375 | 2 | Similar | Nov. 2020 (No) | Langis et al. (2025) |
| AT 2021aeuk | Active | 1300 | 2/3 | Lower (+precursor) | Oct. 2026 | Sun et al. (2025) |
| AT 2023uqm | Passive | 525 | 5 | Higher | Aug. 2026 | Wang et al. (2025) |
| †AT 2024gzn | Active | 400 | 2 | Lower | Sep. 2026 | This work |
| †AT 2023szj | Active | 600 | 2 | Lower | Feb. 2027 | This work |
| ZTF23abjvojy | Active | ~ 5785 | 2 | Lower | 2039 | This work |
| AT 2023adr | Passive | 400 | 2 | Lower | Apr. 2025 (Gap) | This work |




Candidate at the time that the second peak was reported.

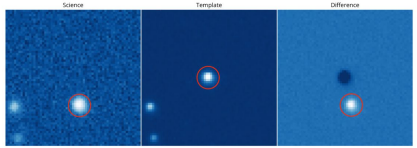
**Spectroscopically confirmed** by Angus *et al.*, 2026, MNRAS, Volume 550, Issue 4

# Case study: Anomaly detection

26 statistical features using `light_curve`
Isolation Forest algorithm
Reporting 10 most anomalous candidates every day, for 2 years

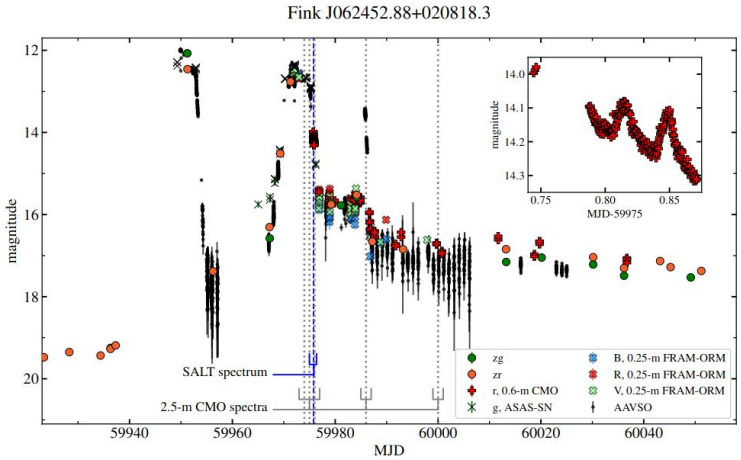
AM CVn (helium CV) of WZ Sge type of variability

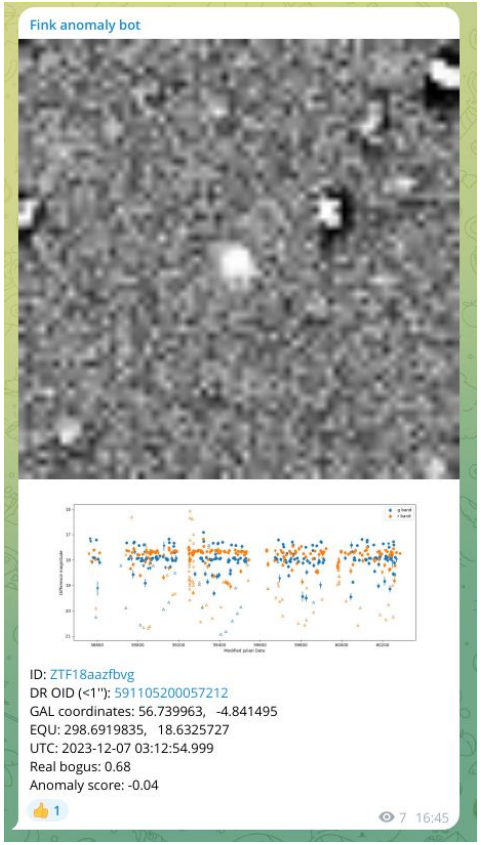
High velocity stars



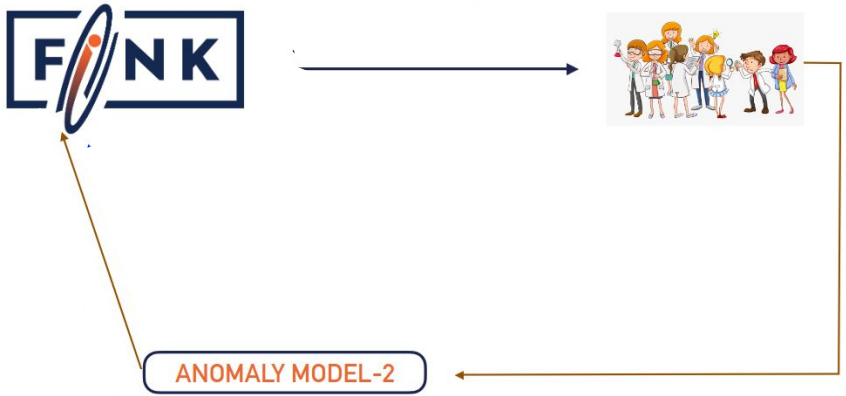

## Fink AD

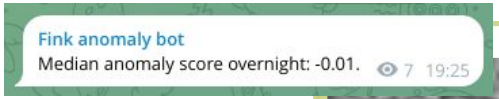
Pruzhinskaya et al., 2026

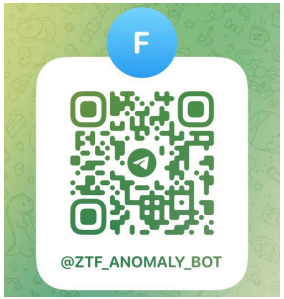


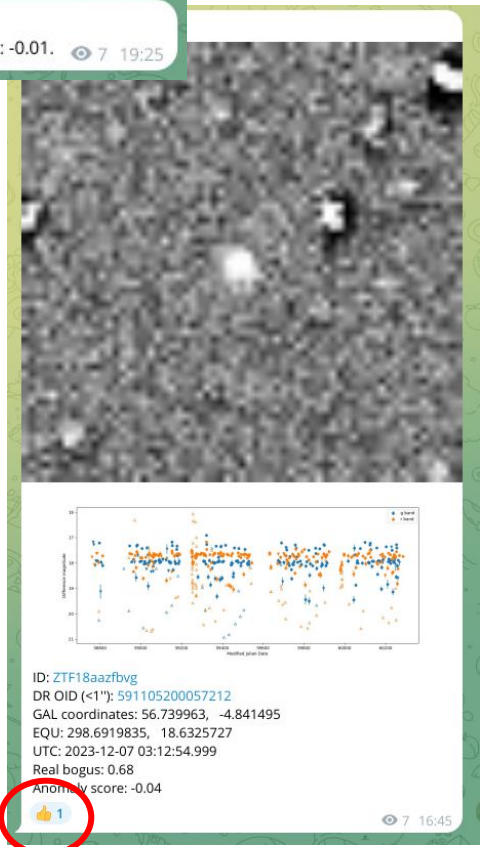



# Discovery in real time



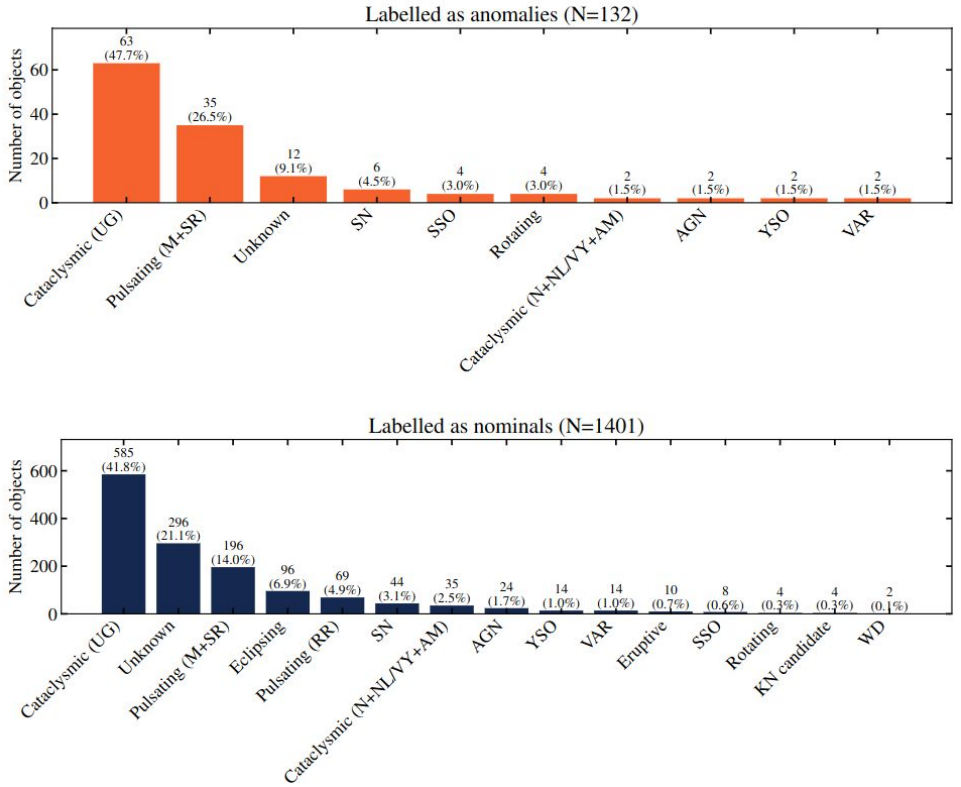




*Algorithm from Das, S., et al., 2017, in DEA'17, KDD workshop, arXiv:cs.LG/1708.09441*
*Implementation by SNAD, via coniferest package (Kornilov et al, 2025 - https://arxiv.org/abs/2410.17142 )*
*Results described in Pruzhinskaya et al., 2026 - https://arxiv.org/abs/2603.29511*

# If only it was that simple …

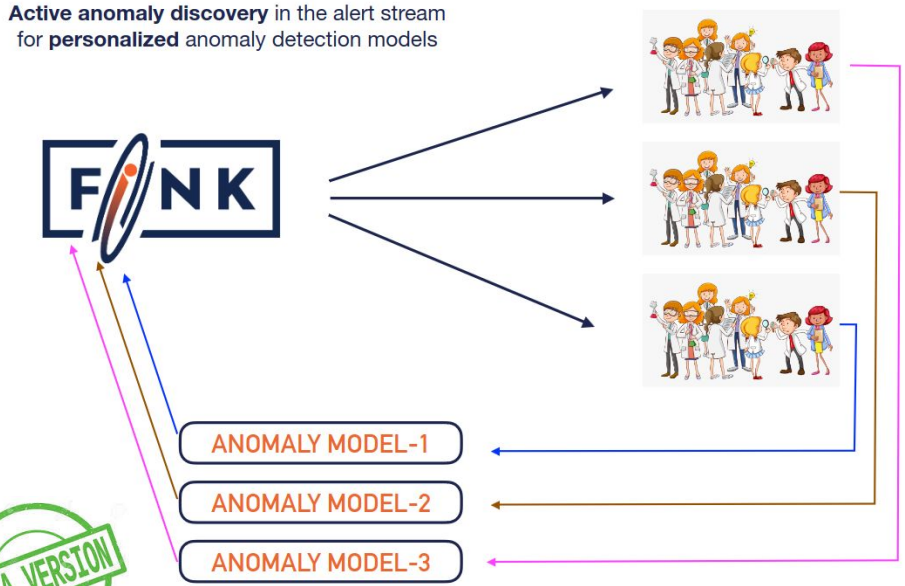
Lessons learned:

- People get bored very easily
- Different people want different things
- They change their minds all the time




*Pruzhinskaya et al., 2026 - https://arxiv.org/abs/2603.29511*

# Personalized models

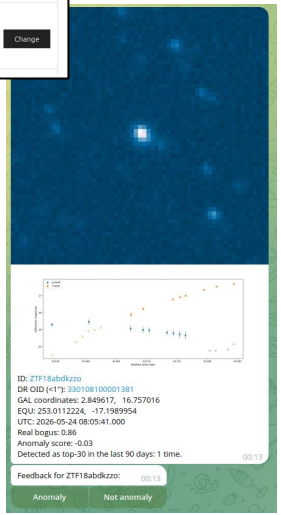



## Your reactions

Hello, emille_30days! You have 733 reactions now. The table content is updated when the page is refreshed.

| candid_id | tag | user | changed_at | | |
|---|---|---|---|---|---|
| 3245412635615015013 (ZTF18acnahlv) | NOT ANOMALY | emille_30days | 2025-11-12 04:50:26 | Delete | Change |
| 3120498706215015001 (ZTF25aalrzkm) | NOT ANOMALY | emille_30days | 2025-08-06 08:35:54 | Delete | Change |
| 3146447334215010004 (ZTF19aabhial) | NOT ANOMALY | emille_30days | 2025-08-06 08:35:53 | Delete | Change |

*Personalized, user friendly ML at scale*



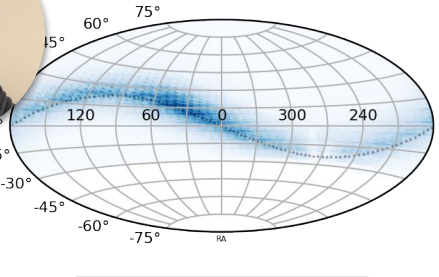

*Algorithm from Das, S., et al., 2017, in DEA'17, KDD workshop, arXiv:cs.LG/1708.09441*
*Implementation by SNAD, via coniferest package (Kornilov et al, 2025 - https://arxiv.org/abs/2410.17142 )*
*Results described in Pruzhinskaya et al., 2026 - https://arxiv.org/abs/2603.29511*

# Example: Solar System Objects

- Table of aggregated parameters for SSO (à la SSObject for LSST): 500k objects (130k@50+ observations)
- Search for new small bodies (linking & fitting) with **Fink-FAT**
- Extending phase function model to account for the variable geometry of the small bodies (spin + oblateness): ***sHG1G2***
- Detecting activation in real-time
- Combining data from multiple surveys: ***collaboration with Euclid SSO WG***

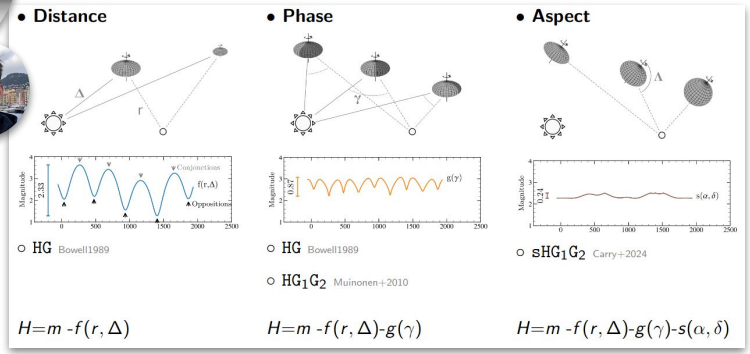
**Enabling discovery of S System object large alert data streams**








**Combined spin orientation and phase function of asteroids, *Carry et al.* 2024**



# Art ⟶ Science ⟶ Art

Life continues to imitate art

Carry *et al.*, 2024, *Combined spin orientation and phase function of asteroids*, A&A,

https://arxiv.org/abs/2403.20179

## Appendix A: Good bet, Mr Clarke!

"*On Day 86 they were due to make their closest approach to any known asteroid, It had no name - merely the number 7794 [...] Through the high-powered telescope, they could see that the asteroid was very irregular, and turning slowly end over end. Sometimes it looked like a flattened sphere, sometimes it resembled a roughly shaped block; [...]*" - *Arthur C. Clarke, 2001 – A Space Odyssey*

In the 1968 Arthur C. Clarke book, *2001 – A Space Odyssey*, the author describes the observation of the asteroid 7794 from a flyby (which would be discovered only in 1996[13]), predicting that this asteroid *"looked like a flattened sphere"*. Using the most recent data from FINK (version `2024.01`, see Sect. 6), we found a value for the oblateness around 0.5, favouring the prediction made in the book. The parameter estimate has still a large uncertainty though, but given the residuals in the phase curve, we can infer that either the asteroid is irregular along its equator ($a \neq b$) or there could be an extra unmodelled activity.

# First Rubin alerts

1:30am (Paris) / 11:30am (Melbourne)
Live monitoring and screening of first alerts
Follow up with Siding Spring

Median Rubin delay ~ 260 seconds
Median Fink delay ~ 70 seconds





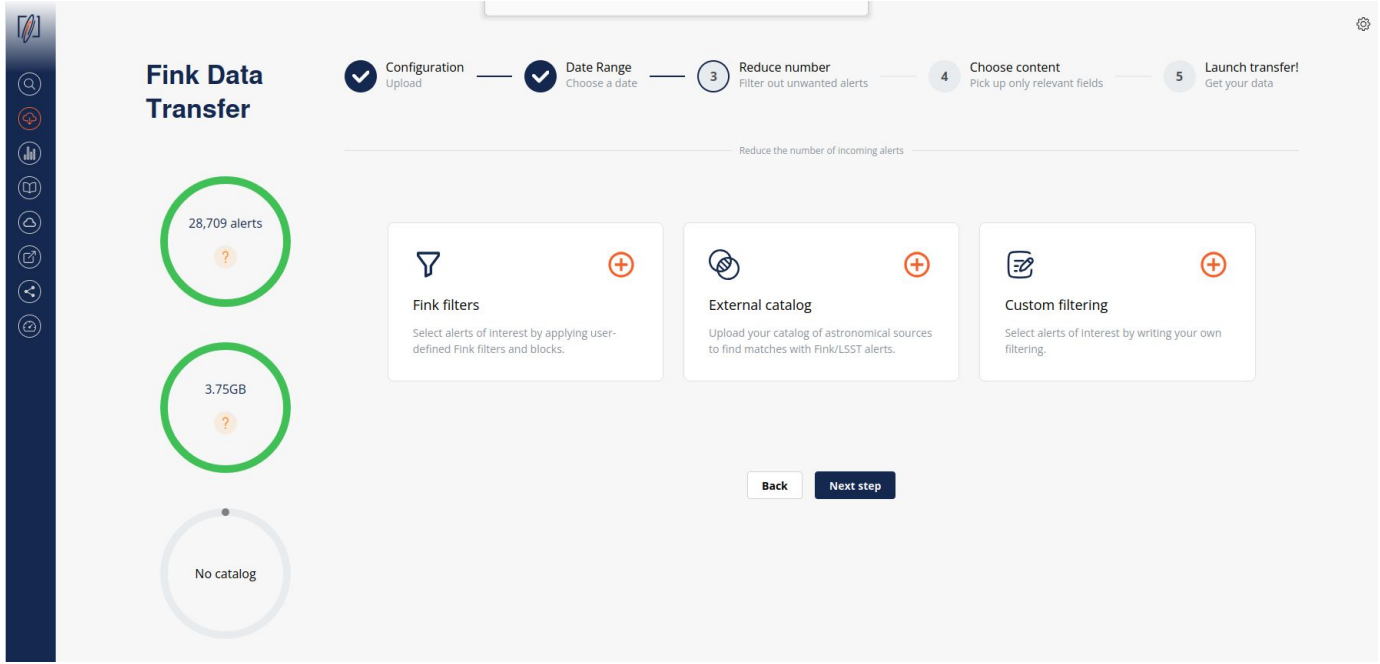


https://fink-broker.org/news/2026-02-26-first-alerts/

# Fink services for Rubin

*Data Transfer:*

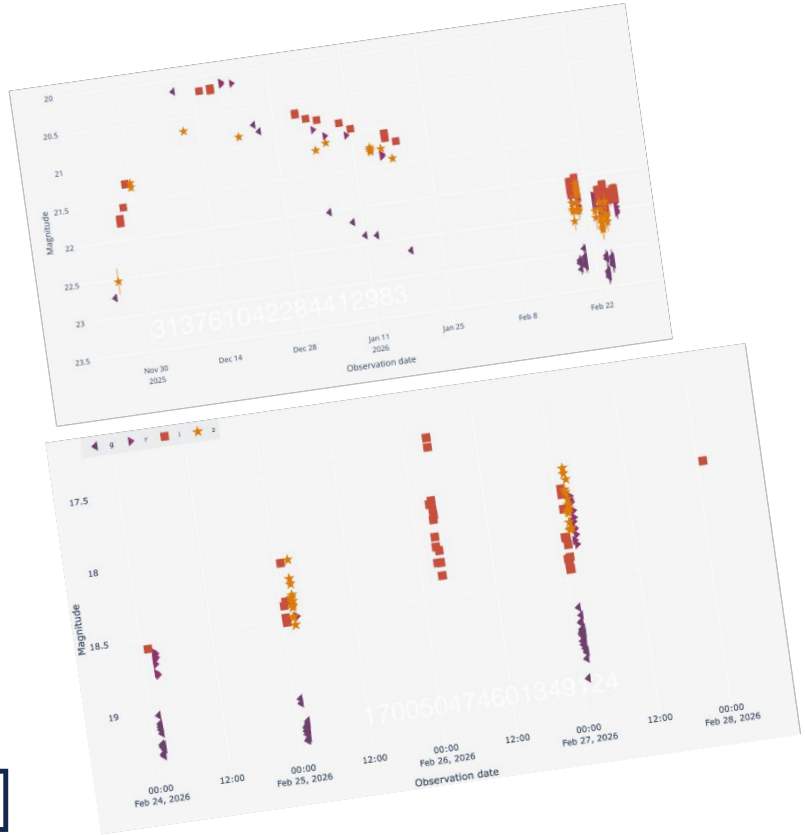
*https://lsst.fink-portal.org/download*

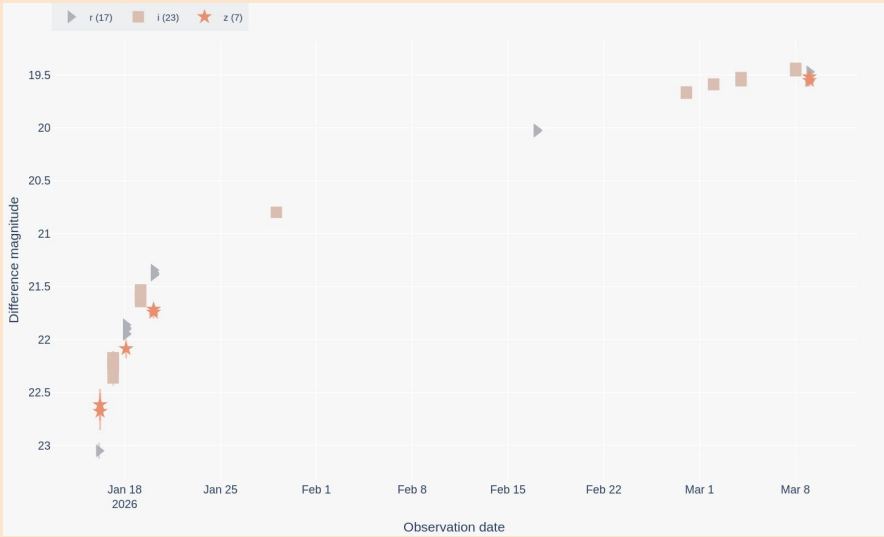



**16 million**
alerts received

**815 million**
downloaded

# Exciting new data





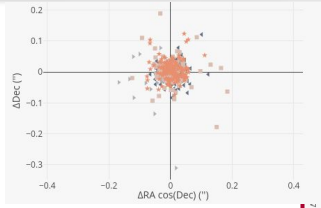


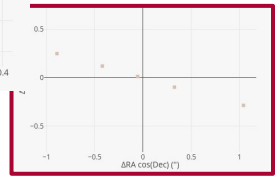
SN IIn
Reported 28 Feb 2026
Classified 03 Mar 2026

https://lsst.fink-portal.org/313994139532263447

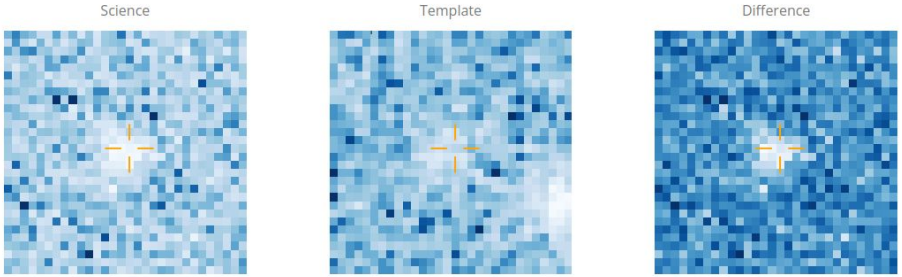
# Search for hostless transients

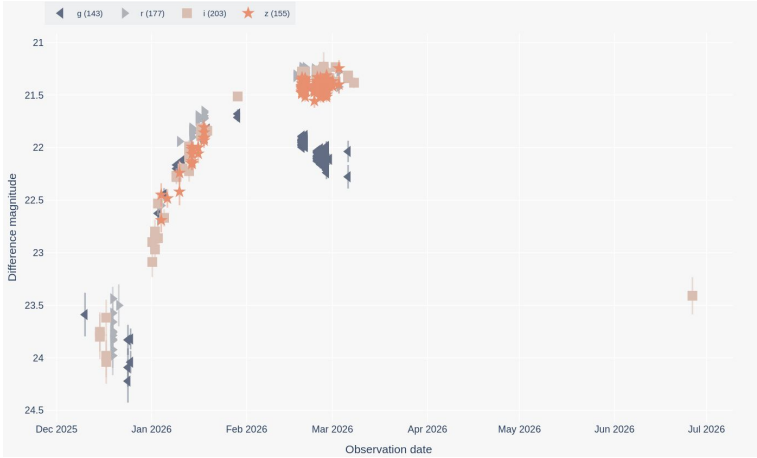
Fined tuned on real ZTF and LSST stamps
Other Fink ML classifications + xmatch + position scatter




**ELEPHANT: hostless extragalactic**
*Pessi et al., 2025*
*Durgesh et al. 2026*






SLSN-I at z=0.96
*Reported as hostless on 26 Feb 2026*
*Classified by Villar et al. on 14 Mar 2026*

*https://lsst.fink-portal.org/313831458352922664*

# For tomorrow

- Build a virtual environment
- Install the fink-client
- Register to get credentials

Detailed instructions here (and in the document Sarang sent)

But most important:

## Be creative!

- Choose a class or an object -- something you want to find
- Think about a list of selection cuts (number of points, duration of the light curves, trends, colors, etc)

# It takes a village

Fink-Brazil Workshop 2024




Fink Collaboration meeting 2024




Fink collaboration meeting 2025




ozFink meeting 2023




Fink hackathon 2022




- *150+ members from 26 countries*
- *High fraction of members from the outside LSST collaborations*
- **Fink@World, June/2026**




https://indico.in2p3.fr/event/38946/

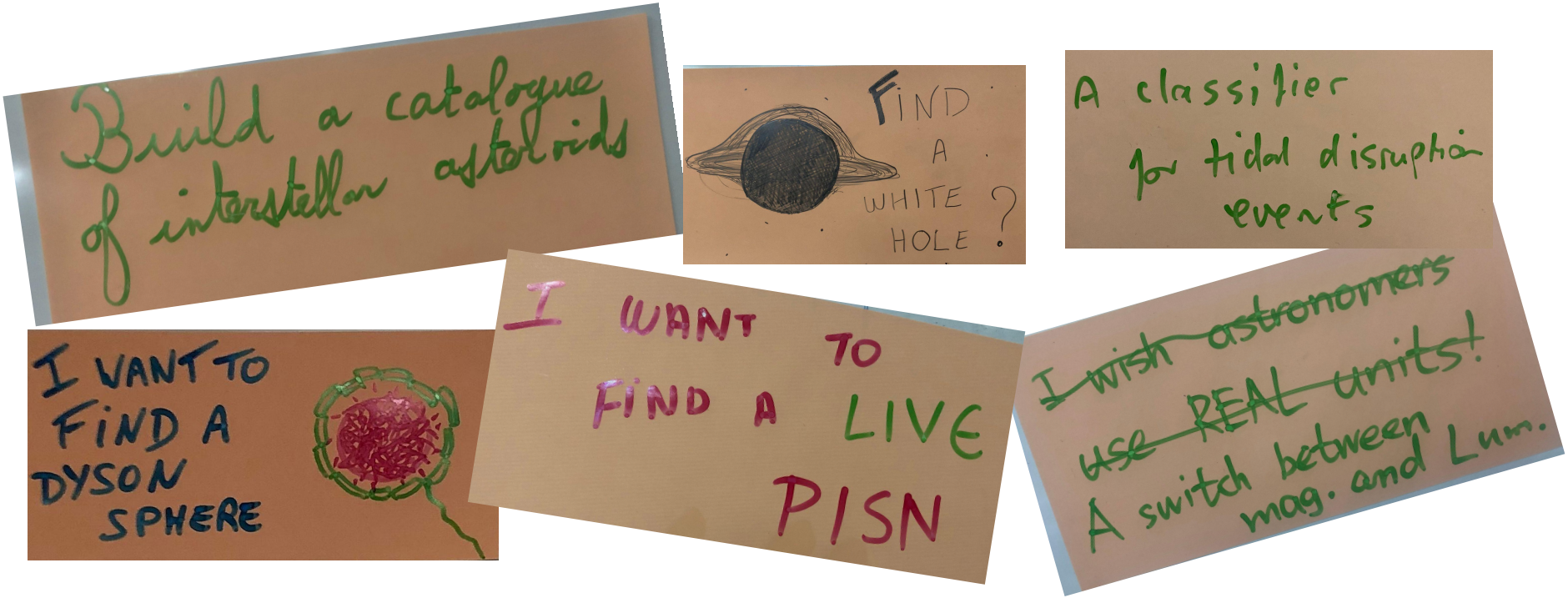



Fink collaboration meeting 2022

ozFink meeting 2025



# #FinkDreamShots

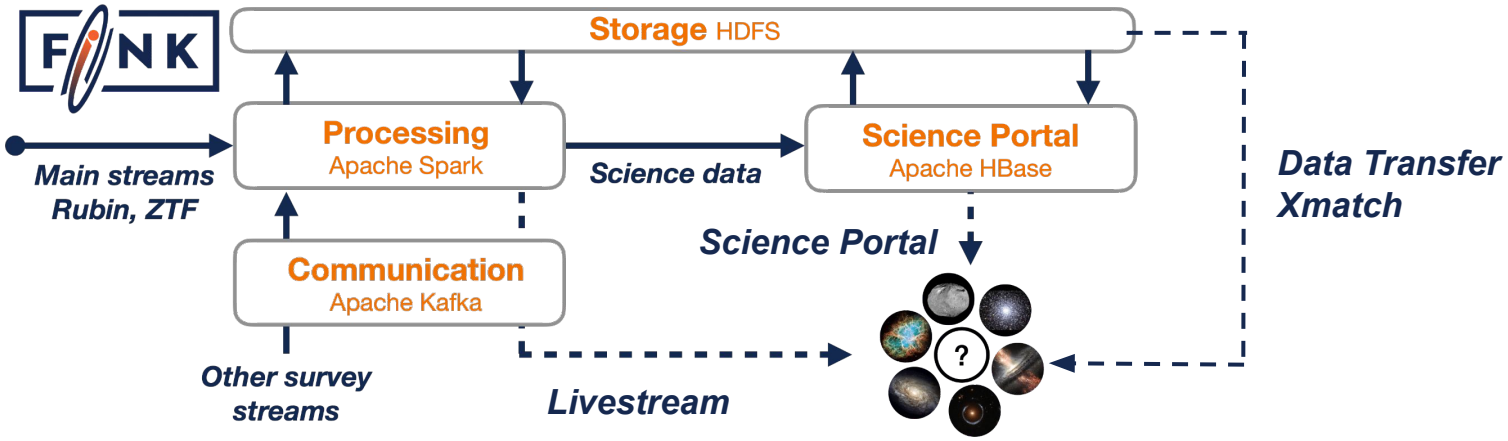



From OzFink 2023 - Melbourne, Australia - [https://www.ozgrav.org/ozfink-workshop-2023.html](https://www.ozgrav.org/ozfink-workshop-2023.html)

# What do you want to see?



# High level architecture



# High level architecture

